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Course Code: CSA1402

Compiler Design Lab Programs

18. Write a LEX program to count the number of Macros defined and header files included in the C program.

Code:

```
%{  
#include <stdio.h>  
  
int macros=0, headers=0;  
  
%}  
%%  
^[ \t]*#define\b { macros++; }  
^[ \t]*#include\b { headers++; }  
.\n ;  
%%  
  
int main(){ yylex(); printf("Macros: %d\nHeaders: %d\n", macros, headers); return 0; }  
int yywrap(){ return 1; }
```

Output:

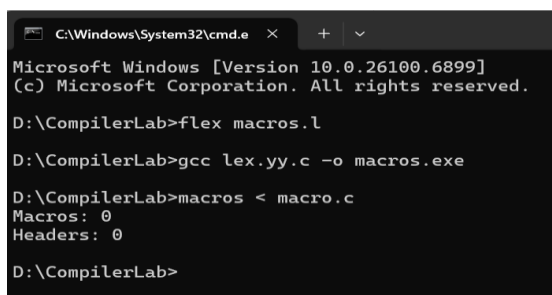
```
#define PI 3.14  
#include <stdio.h>  
#include <conio.h>  
  
makefile
```

Copy code

Macros: 1

Headers: 2

Output Screenshot:



```
C:\Windows\System32\cmd.e  x  +  v  
Microsoft Windows [Version 10.0.26100.6899]  
(c) Microsoft Corporation. All rights reserved.  
D:\CompilerLab>flex macros.l  
D:\CompilerLab>gcc lex.yy.c -o macros.exe  
D:\CompilerLab>macros < macro.c  
Macros: 0  
Headers: 0  
D:\CompilerLab>
```

19. Write a LEX program to print all HTML tags in the input file.

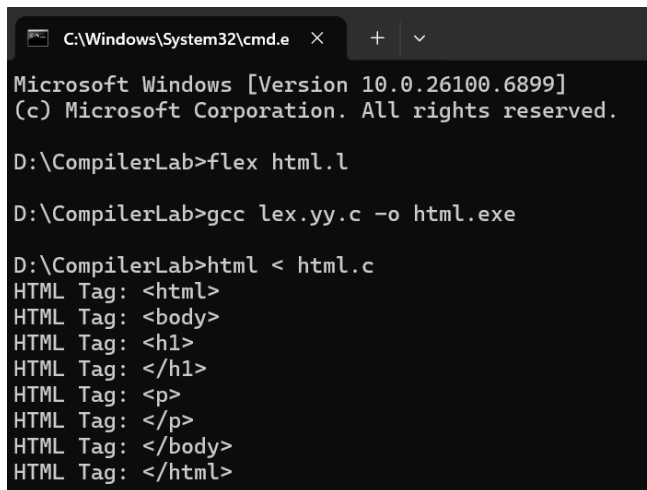
Code:

```
%{  
#include <stdio.h>  
%}  
%%  
\<[^>]+\>  { printf("HTML Tag: %s\n", yytext); }  
.\n        ;  
%%  
  
int main(){ yylex(); return 0; }  
int yywrap(){ return 1; }
```

Output:

```
<html>  
<body>  
<h1>My First Heading</h1>  
<p>My first paragraph.</p>  
</body>  
</html>  
  
HTML Tag: <html>  
HTML Tag: <body>  
HTML Tag: <h1>  
HTML Tag: </h1>  
HTML Tag: <p>  
HTML Tag: </p>  
HTML Tag: </body>  
HTML Tag: </html>
```

Output Screenshot:



```
C:\Windows\System32\cmd.e  X  +  v
Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

D:\CompilerLab>flex html.l

D:\CompilerLab>gcc lex.yy.c -o html.exe

D:\CompilerLab>html < html.c
HTML Tag: <html>
HTML Tag: <body>
HTML Tag: <h1>
HTML Tag: </h1>
HTML Tag: <p>
HTML Tag: </p>
HTML Tag: </body>
HTML Tag: </html>
```

20. Write a LEX program which adds line numbers to the given C program file and display the same in the standard output.

Code:

```
%{
#include <stdio.h>

int lineno = 1;

%}

%%

[^\n]*\n  { printf("%4d: %s", lineno++, yytext); }
[^\n]+    { printf("%4d: %s\n", lineno++, yytext); }

%%

int main(){ yylex(); return 0; }

int yywrap(){ return 1; }
```

Output:

```
#define PI 3.14
#include<stdio.h>

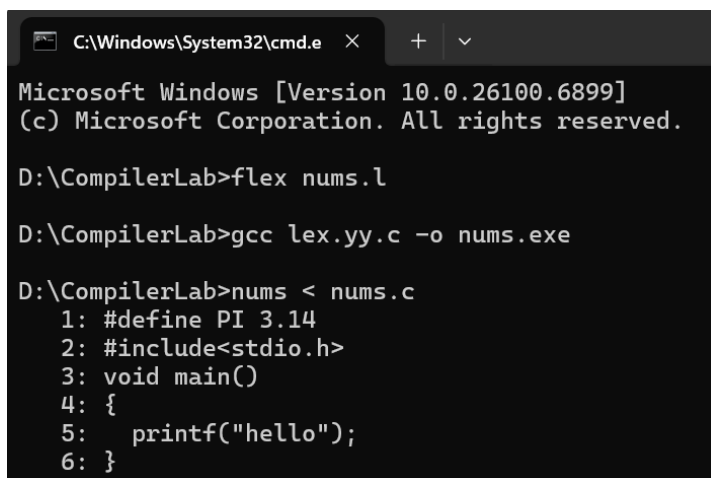
void main()
{
    printf("hello");
}

cpp
```

Copy code

```
1: #define PI 3.14
2: #include<stdio.h>
3:
4: void main()
5: {
6:   printf("hello");
7: }
```

Output Screenshot:



```
C:\Windows\System32\cmd.e  X  +  v
Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

D:\CompilerLab>flex nums.l

D:\CompilerLab>gcc lex.yy.c -o nums.exe

D:\CompilerLab>nums < nums.c
1: #define PI 3.14
2: #include<stdio.h>
3: void main()
4: {
5:   printf("hello");
6: }
```

21. Write a LEX specification count the number of characters, number of lines & number of words.

Code:

```
%{
#include <stdio.h>

int chars=0, words=0, lines=0;
%}
%%
[ \t]+      ;
\n          { lines++; chars++; }
[A-Za-z0-9_]+{ words++; chars += yyleng; }
.           { chars += yyleng; }
%%
```

```
int main(){ yylex(); printf("Chars:%d Words:%d Lines:%d\n", chars, words, lines); return 0;
}
```

```
int yywrap(){ return 1; }
```

Output:

Hello world

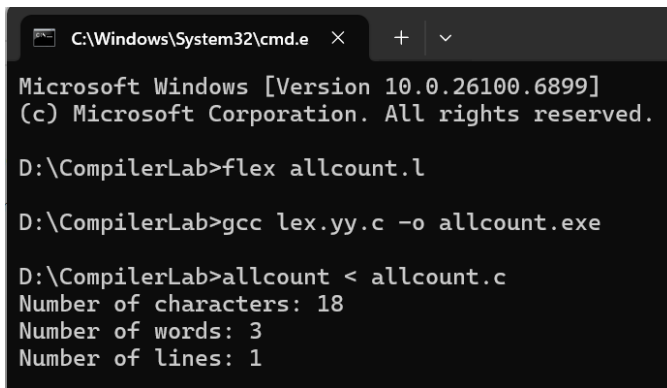
Welcome

Chars:19

Words:3

Lines:2

Output Screenshot:



```
C:\Windows\System32\cmd.e  X  +  v
Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

D:\CompilerLab>flex allcount.l

D:\CompilerLab>gcc lex.yy.c -o allcount.exe

D:\CompilerLab>allcount < allcount.c
Number of characters: 18
Number of words: 3
Number of lines: 1
```

22. Write a LEX program to count the number of comment lines in a given C program and eliminate them and write into another file.

Code:

```
%{
#include <stdio.h>

int comment_lines=0;

FILE *out;

%}

%%

"/*".*      { comment_lines++; }

"/*"([^\*]|\\*+[/])*\*/"  { /* attempt to match block comment */ comment_lines++; }

[^\n]+      { fprintf(out, "%s", yytext); }

\n          { fprintf(out, "\n"); }

%%
```

```

int main(){
    yyin = stdin;
    out = fopen("output.c","w");
    if(!out){ perror("output.c"); return 1; }
    yylex();
    fclose(out);
    printf("Comment lines: %d\nNon-comment code written to output.c\n", comment_lines);
    return 0;
}
int yywrap(){ return 1; }

```

Output:

```
#include<stdio.h>
```

```

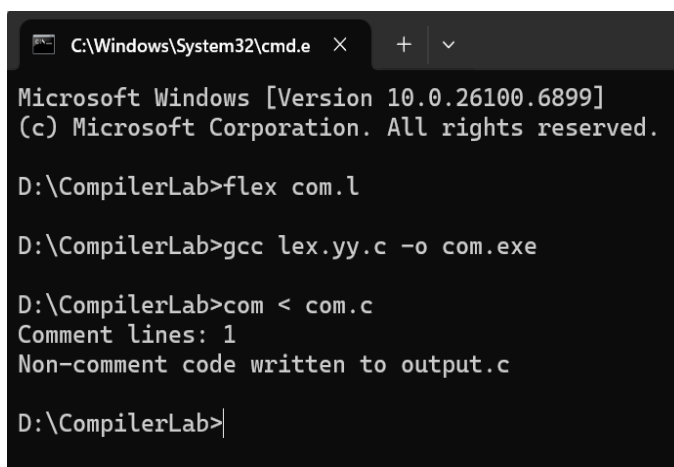
int main()
{
    int a; /*var declaration*/
    printf("hi");
    return 0;
}

```

Comment lines: 2

Non-comment code written to output.c

Output Screenshot:



```

C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

D:\CompilerLab>flex com.l

D:\CompilerLab>gcc lex.yy.c -o com.exe

D:\CompilerLab>com < com.c
Comment lines: 1
Non-comment code written to output.c

D:\CompilerLab>

```